

PDF Parsing Library Comparison

Resume Parsing Project — Prepared for Mahnoor

For this task, I installed and tested three Python libraries used for extracting text from PDF files — PyMuPDF, pdfplumber, and pdfminer.six — and evaluated each one against seven criteria: installation, ease of use, speed, accuracy, preservation of formatting, advantages, and limitations. All three were tested on the same real resume PDF so the comparison would be based on actual results rather than documentation alone.

1. Installation

PyMuPDF (fitz)	pdfplumber	pdfminer.six
Installed via pip, but failed to import at first with a DLL error. Needed the Microsoft Visual C++ Redistributable installed and the computer restarted before it worked.	Installed via pip and worked immediately, no issues.	Installed via pip and worked immediately, no issues.

2. Ease of Use

PyMuPDF (fitz)	pdfplumber	pdfminer.six
Simple API — just a couple of lines to open a PDF and pull out text (pymupdf.open, get_text()).	Simple API — pdfplumber.open, extract_text(). Also has built-in table extraction if needed.	Simple function call (extract_text()), fewer configuration options than the other two.

3. Speed

PyMuPDF (fitz)	pdfplumber	pdfminer.six
Fastest of the three — noticeably quicker to return results.	Moderate — a bit slower than PyMuPDF, but still fast enough for a single resume.	Generally the slowest of the three, especially on larger files.

4. Accuracy

PyMuPDF (fitz)	pdfplumber	pdfminer.six
All the correct words were extracted — nothing was missing from the text itself.	All the correct words were extracted — nothing was missing from the text itself.	All the correct words were extracted, but one stray symbol appeared in the output that isn't present anywhere in the real resume.

5. Preservation of Formatting

PyMuPDF (fitz)	pdfplumber	pdfminer.six
Mostly preserved, but moved the header text ('Resume: Digital Arts / Font: Verdana') from the top of the page to the very end of the output, and separated job titles from their dates onto different lines.	Best of the three — job titles stayed on the same line as their dates, and the header stayed at the top, matching the original page exactly.	Weakest of the three — grouped all job titles together first, then all dates in a separate block afterward, breaking the original reading order. Also left extra blank lines throughout.

6. Advantages

PyMuPDF (fitz)	pdfplumber	pdfminer.six
Very fast; documentation suggests it also handles images and complex PDFs well.	Best formatting/reading-order accuracy; built-in table extraction could help with resumes that use tables for skills or education.	Mature and widely used library; works fine as a fallback option.

7. Limitations

PyMuPDF (fitz)	pdfplumber	pdfminer.six
Needs the Visual C++ Redistributable installed on Windows — an extra setup step beyond a plain pip install.	Slightly slower than PyMuPDF on very large files — not a real issue for single resumes.	Reading order can get scrambled, particularly on documents with more complex layouts.

Test Method

I installed all three libraries in the same Python setup, then ran each one on page 3 of a sample resume PDF downloaded from a university career center (a Digital Arts resume). I then compared each library's raw text output against how the resume actually looks, to check whether the text came out in the right order and whether anything was missing.

Recommendation

Based on this comparison, I'm using **pdfplumber** for the text extraction module in Task 3. It had no installation issues and gave the most accurate, correctly ordered output of the three — nothing was shuffled or separated from where it belonged in the original resume. PyMuPDF is a strong second choice if speed becomes important on a larger batch of resumes, though it requires the extra Visual C++ Redistributable setup step. pdfminer.six can be kept as a fallback option, but its output needs more cleanup due to the scrambled reading order observed

during testing.

Appendix: Sample Extraction Output (Page 3, Digital Arts Resume)

PyMuPDF output (excerpt):

```
Graphic Designer
August 2019 - Present
Freelance Web & Graphic Design
...
SKILLS
...
Resume: Digital Arts
Font: Verdana
(header moved from top of page to the end of the output)
```

pdfplumber output (excerpt):

```
Resume: Digital Arts
Font: Verdana
Allie Gator
Gainesville, FL 32611 | 352-290-5678 | albertagator@ufl.edu
EDUCATION
Bachelor of Arts in Digital Arts June 2023
PROFESSIONAL EXPERIENCE
Graphic Designer August 2019 - Present
(header stayed at top; title and date stayed on one line, exact match to the original)
```

pdfminer.six output (excerpt):

```
Graphic Designer
Freelance Web & Graphic Design
Studio Co-Founder & Leader
Design Awaken Arts, Gainesville, FL
June 2023
August 2019 - Present
May 2020 - Present
(titles grouped first, then all dates dumped separately afterward - reading order broken)
```